

## PL/SQL Exercise 1: Control Structures

The screenshot shows the Oracle FreeSQL interface. The main editor displays the following PL/SQL code:

```
1 SET SERVEROUTPUT ON;
2
3 -- Scenario 1: Apply 1% loan interest discount to customers over 60
4 DECLARE
5     v_age NUMBER;
6 BEGIN
7     FOR cust IN (SELECT CustomerID, Name, DOB FROM Customers) LOOP
8         v_age := TRUNC(MONTHS_BETWEEN(SYSDATE, cust.DOB) / 12);
9
10        IF v_age > 60 THEN
11            UPDATE Loans
12            SET InterestRate = InterestRate - (InterestRate * 0.01)
13            WHERE CustomerID = cust.CustomerID;
14
15            DBMS_OUTPUT.PUT_LINE(cust.Name || ' (age ' || v_age || ')');
16        END IF;
17    END LOOP;
18    COMMIT;
19 END;
```

Below the code, the following SQL statement is visible:

```
22 -- Scenario 2: Flag customers with balance > $10,000 as VIP
23 ALTER TABLE Customers ADD IsVIP VARCHAR2(5) DEFAULT 'FALSE';
```

The left sidebar shows a Navigator with a tree view of the database schema, including tables like ACCOUNTS, CUSTOMERS, EMPLOYEES, LOANS, and TRANSACTIONS. The right sidebar displays SQL Tutorials, including "Creating Tables: Databases fo..." and "Joining Tables: Databases for...".

The screenshot shows the Oracle FreeSQL interface after executing the PL/SQL code. The main editor displays the following PL/SQL code:

```
6 BEGIN
7     FOR cust IN (SELECT CustomerID, Name, DOB FROM Customers) LOOP
8         v_age := TRUNC(MONTHS_BETWEEN(SYSDATE, cust.DOB) / 12);
9
10        IF v_age > 60 THEN
```

Below the code, the execution results are displayed in a table with columns: Query result, Script output, DBMS output, Explain Plan, and SQL history. The results are as follows:

| Query result | Script output | DBMS output                                                  | Explain Plan | SQL history |
|--------------|---------------|--------------------------------------------------------------|--------------|-------------|
|              |               | John Doe (age 66) got a 1% loan discount.                    |              |             |
|              |               | John Doe promoted to VIP (balance: 12000)                    |              |             |
|              |               | REMINDER: Jane Smith, your loan (ID 2) is due on 06-AUG-2026 |              |             |

The left sidebar shows the same Navigator as the previous screenshot. The right sidebar displays SQL Tutorials, including "Creating Tables: Databases fo..." and "Joining Tables: Databases for...".